



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course: CSE 489

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Project Title: Advertising and Printing Business App

Project Features:

This is an Android application built with Flutter and Dart for a small printing and advertising shop. It lets a customer see the services, look at previous work, contact the shop in several ways and ask for a price quotation. Saved data is kept in a local SQLite database on the phone. By the end of this course, the following features were implemented:

- a) Splash screen that shows the company logo and then opens the home page by itself.
- b) Home page with a company banner, a short about section, quick contact buttons and a few popular services.
- c) Services page that lists every service, grouped into printing services and advertising services.
- d) Gallery page that shows previous work in a two column grid, with a full screen viewer that can be zoomed with two fingers.
- e) Contact page that shows the company address, phone number, email address and business hours.
- f) Call Now button that opens the phone dialer with the business number already typed in.
- g) WhatsApp button that opens a chat with the shop, and shows a clear message if WhatsApp is not installed.
- h) Email button that opens the email app with the business address and a subject filled in.
- i) Open in Google Maps button that shows the shop location in the maps app.
- j) Social media buttons for Facebook, Instagram and YouTube.
- k) Share App button that opens the normal Android share menu with a short message about the shop.
- l) Request Quote form with validation for the name, phone number, email address, service, quantity and description.
- m) Quote requests are saved in a local SQLite database, so they are still there after the app is closed and opened again.
- n) My Requests page where a saved request can be read, marked as New, Sent or Done, deleted, or sent to the shop over WhatsApp.

- o) Service Helper chatbot. It first tries the free Google Gemini API for a natural reply. If there is no internet or no API key, it answers from a built in list of rules instead, so it always works.
- p) Every chatbot question and answer is stored in the database, together with a note saying whether the online AI or the offline rules answered it.
- q) About page that explains the shop, the app version, which chatbot mode is active and where the customer data is stored.
- r) Bottom navigation bar and a side menu for moving between all the pages.
- s) Unit tests and widget tests for the chatbot rules, the data models and the main screens.

Online Resources used:

a) Reference:

- *Flutter official documentation at docs.flutter.dev, mainly the pages on widgets, navigation and forms.*
- *Dart language documentation at dart.dev for null safety and async programming.*
- *Package pages on pub.dev for url_launcher, share_plus, sqflite, path and http.*
- *Google Gemini API documentation at ai.google.dev for the generateContent endpoint and the free tier.*
- *Android developer documentation on package visibility, which explains the queries tag needed on Android 11 and above.*
- *SQLite documentation at sqlite.org for the table and query syntax.*
- *YouTube tutorials on Flutter basics, bottom navigation and using the sqflite package.*

b) Stack Overflow and GitHub links:

- *Project repository: <https://github.com/siffyyrox-x/Advertising-and-Printing-Business-App>*
- *Stack Overflow answers on opening WhatsApp from a Flutter app with url_launcher.*
- *Stack Overflow answers on the queries tag in AndroidManifest.xml, which is needed before an app can open the dialer or WhatsApp on newer Android versions.*
- *Stack Overflow answers on using FutureBuilder together with sqflite to show database rows in a list.*
- *GitHub example projects and README files of the url_launcher, share_plus and sqflite packages.*

Future Enhancements:

The application is working, but it is still a first version. The following enhancements can be added later to make the system more complete and more useful for the shop:

- a) An owner side of the app, so the shop can add new services and gallery pictures without changing the code.
- b) Sending a quote request straight to the shop through a small server or an email service, instead of the customer sending it over WhatsApp.
- c) A simple login system, so a customer can see their old requests on any phone.
- d) Push notifications to tell the customer as soon as the shop replies with a price.
- e) Order tracking, so the customer can see whether the job is being printed, is ready, or has been delivered.
- f) An option to attach a design file or a picture along with the quote request.

- g) Online payment or an advance booking option for confirmed orders.
- h) Bangla language support, because many local customers are more comfortable reading Bangla.
- i) A dark theme, so the app is easier on the eyes at night.
- j)** Saving the gallery pictures on a server, so new work can be shown without publishing a new version of the app.